

Dr. Haytham Ashraf Abdelaal Ali

JSPS Postdoctoral Research Fellow – University of Tsukuba, Japan

Email: haytham_ali@science.sohag.edu.eg

ORCID: 0000-0002-7623-0887 Google Scholar: Haytham A. Ali

LinkedIn: Haytham Ashraf GitHub: github.com/Haytham-ali

Research Interests

Researcher in AI for Medical Imaging with expertise in physics-guided deep learning, generative modeling, and computational inverse problems. My work focuses on developing robust and interpretable AI systems for volumetric medical imaging, integrating physics-based modeling with data-driven learning. I am particularly interested in next-generation multimodal and prompt-based frameworks for 3D image reconstruction, segmentation, and clinically deployable decision-support systems. My research emphasizes reliability, generalization, and translational impact in real-world clinical environments.

Keywords: Generative models, 3D medical image segmentation, Multimodal learning, Inverse problems, Computational imaging, Clinical AI.

Work Experience

JSPS Postdoctoral Research Fellow Systems and Information Engineering University of Tsukuba, Japan	07/2024–Present
Lecturer Mathematics Department, Faculty of Science Sohag University, Egypt	03/2024– 06/2024
Assistant Lecturer Mathematics Department, Faculty of Science Sohag University, Egypt	12/2023– 03/2024
Teaching Assistant Systems and Information Engineering University of Tsukuba, Japan	07/2021–03/2023
Assistant Lecturer Mathematics Department, Faculty of Science Sohag University, Egypt	11/2018– 09/2020
Research Assistant Mathematics Department, Faculty of Science Sohag University, Egypt	10/2014– 10/2018

Education

PhD in Engineering Systems and Information Engineering, University of Tsukuba, Japan Thesis: <i>Parametric Level-Set Method for Geometric Tomography Reconstruction</i>	10/2020–09/2023
Master of Science in Mathematics Faculty of Science, Sohag University, Egypt	04/2016–09/2018
Bachelor of Science Graduated with Distinction (Excellent with Honors)	09/2009–07/2013

Technical Skills

- **Deep Learning:** PyTorch; generative and attention-based models for CT reconstruction and inverse problems.
- **Imaging & Inverse Problems:** CT reconstruction (FBP and iterative methods), forward/adjoint modeling; ASTRA Toolbox, Torch Radon.
- **Programming:** Python, MATLAB, Mathematica.

Grants & Awards

Competitive Fellowships & Major Grants

- **JSPS Research Fellowship**, Prestigious fellowship awarded by the Japan Society for the Promotion of Science (JSPS) for research excellence 07/2024–07/2026
- **Grant-in-Aid for JSPS Fellows**, (KAKENHI Project Number: 24KF0101), University of Tsukuba, Japan, Co-Investigator 07/2024–03/2027
- **NEC C&C Foundation Research Grant**, University of Tsukuba, Japan, Principal Investigator 10/2022–03/2023
- **Monbukagakusho (MEXT) Scholarship**, Ministry of Education, Culture, Sports, Science and Technology (MEXT), Japan 10/2020–09/2023
- **Science & Technology Development Fund (STDF)**, Sohag University, Egypt, Co-Principal Investigator 04/2020–03/2021

Awards & Honors

- **Best Presentation Award**, International Conference on Computer and Automation Engineering (ICCAE), Perth, Australia, 2025
- **Excellent Oral Presentation Award**, ICCAE, Sydney, Australia, 2023
- **Academic Group Chair Award**, for outstanding research achievements during my Ph.D., University of Tsukuba, Japan, 2023

Travel & Other Grants

- **Travel Grant**, RIMS Workshop on Inverse Problems, Medical Imaging, and Related Topics, Kyoto, Japan, 2023

Selected Publications

Refereed Journal Articles

1. PIDA-GAN: Physics-Informed Dual-Stage Attention GAN with Data Consistency for Sparse-View CT Reconstruction
Haytham A. Ali, Hiroyuki Kudo
Under Review, 2026.
2. Compressed Sensing-Based Image Reconstruction for Discrete Tomography with Sparse View and Limited Angle Geometries
Haytham A. Ali, Essam Rashed, Hiroyuki Kudo
PLOS ONE, 2025.

3. Binary Tomography Reconstruction with Limited-Data by a Convex Level-Set Method
Haytham A. Ali, Hiroyuki Kudo
CMC-Computers Materials & Continua, 2022.
4. Generating Bézier curves for medical image reconstruction
H. S. Abdel-Aziz, E. A. Zanaty, **Haytham A. Ali**, and M. K. Saad
Results in physics, 2021.

Refereed Conference Papers

1. **Haytham A. Ali**, Hiroyuki Kudo, PCDS-GAN: Physics-Constrained Dual-Stage Generative Model with Adaptive Gating for Ultra Sparse-View CT Reconstruction, **ISBI** 2026 (Accepted).
2. **Haytham A. Ali**, Hiroyuki Kudo, Multi-Frequency Domains with Modified U-Net for Limited-Angle CT, **ISBI** 2026 (Accepted).
3. **Haytham A. Ali**, Hiroyuki Kudo, FLoAT-GAN: Frequency- and Locality-Guided Attention for Limited-Angle CT Reconstruction, **AAAI-26** W3PHIAI Workshop (Long Presentation), 2026.
4. **Haytham A. Ali**, Hiroyuki Kudo, Attention-Guided Frequency and Locality Learning GAN for LACT, **IWAIT/IFMIA**, 2026.
5. **Haytham A. Ali**, Hiroyuki Kudo, An Attention-Based Generative Adversarial Network for Limited-Angle CT Reconstruction, **ICCAE**, 2025.
6. **Haytham A. Ali**, Hiroyuki Kudo, Level-Set Method for Limited-Data Reconstruction in CT using Dictionary-Based Compressed Sensing, **ICCAE**, 2023.
7. **Haytham A. Ali**, Hiroyuki Kudo, New Level-Set-Based Shape Recovery Method and its application to sparse-view shape tomography, **DMIP**, 2021.

Presentations & Invited Talks

1. **Haytham A. Ali**, Hiroyuki Kudo, Novel Dual-Stage GAN for High-Fidelity CT Image Reconstruction from Sparse view CT, Annual Meeting of the Japanese Society for Medical Imaging (JAMIT), 2025.
2. **Haytham A. Ali**, Hiroyuki Kudo, Self-Attention GAN for High-Quality Limited-Angle CT Reconstruction, Annual Meeting of the Japanese Society for Medical Imaging (JAMIT), 2025.
3. **Haytham A. Ali**, Hiroyuki Kudo, Towards Accurate Reconstruction from Sparse and Limited-Angle CT Data Using Deep Learning, Asia Symposium on Image Processing, 2025.
4. **Haytham A. Ali**, Hiroyuki Kudo, Parametric Level Set Methods for Limited Data CT Image Reconstruction, Workshop on Inverse Problems, Medical Imaging, and Related Topics, 2023.
5. **Haytham A. Ali**, Katsuya Fujii, Hiroyuki Kudo, Reconstruction method for binary limited-data tomography using a dictionary-based sparse shape recovery, Annual Meeting of the Japanese Society for Medical Imaging (JAMIT), 2022.

Teaching Experience

Lecturer, Sohag University

03/2024–06/2024

Courses: Calculus, Linear Algebra, Geometry, and Double integrals

Responsibilities: delivery of lectures and tutorials, preparation of course materials and assessments, grading examinations, and providing academic advising and office hours.

Teaching Assistant, Sohag University

12/2023–03/2024

Courses: *Geometry, Differential Geometry, Calculus, Integral Calculus, and Linear Algebra*

Responsibilities: preparation of lecture materials, grading assignments and exams, and guiding students during tutorials and office hours.

Teaching Assistant, University of Tsukuba

07/2021–03/2023

ALD503: Computer Science

ALD504: Research in Computer Science

Responsibilities: organizing and preparing seminars, monitoring attendance, and mentoring students on course activities.

Teaching Assistant / Instructor, Sohag University

04/2016–09/2020

Courses: *Analytic Geometry, Differential Geometry, Calculus, Integral Calculus, Linear Algebra, Statics, and Dynamics*

Responsibilities: preparation of lecture materials, grading assignments and exams, guiding students during tutorials and office hours, and solving exercises with students.

References

- **Prof. Hiroyuki Kudo** 10/2020 – Present
Ph.D. Advisor & Host, University of Tsukuba
kudo@cs.tsukuba.ac.jp
- **Prof. Essam Rashed** 10/2020 – Present
Collaborator & Mentor, University of Hyogo
rashed@gsis.u-hyogo.ac.jp
- **Prof. M. Khalifa Saad** 04/2016 – 10/2020
Advisor, Sohag University
mohamed.khalifa77@science.sohag.edu.eg